Last leg to Block I: corridor, entrance, stop

Nov 8, 2026 | [name to be added] | about 90 seconds

One-minute script

> The last leg starts after the left turn at the end of the arcade and ends at the glass entrance of Block I, the KARL entrance, where the rover has to stop. First it follows the wall, then it crosses the interlock walkway, then it drives between two railings.
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> The walkway has no wall. On its left there is grass with benches and a hydrant, and later the low railing and drain at the foot of the building. The rover must not drift into the grass or mud, so it centres itself on that left edge: the LiDAR fits a line through those objects, the rover remembers the distance it first saw, and steers gently to keep it. The honest limit: in the campus 9 replay this line was found in only about 10 percent of the scans; the rest of the time it holds its heading.
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> In campus 9 the rover did not stop at the entrance and hit a bollard nine metres later. The railing rule was too strict, and the bollard came closer than 35 centimetres, below the LiDAR minimum range.
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> Version 10 recognises the corridor by its width and the entrance where the facade steps back and the glass loses LiDAR points, then stops 1.5 metres later. In replay: corridor at 35.6 metres, entrance at 47.0, stop at 48.5, before the bollard at 56. Replay only; it still has to be driven.

What you need to understand

The goal: stop gently at the right place, not just before an obstacle.

- Phases: wall follow at 1.0 m (power 0.30), walkway (centering on the left edge), railing corridor, KARL entrance, stop. All stops are slow ramps because of the gearbox.
- Walkway centering, in detail: the box fits a straight line through LiDAR points on the left, 1 to 3 m away, at least 5 points over 0.8 m, within ±12 degrees of the heading. The first distance it sees (limited to 1.3 to 2.4 m) is kept. Too far from the edge: steer left, 10 degrees per metre, at most ±6 degrees, changed slowly. The LiDAR sees shapes, not names: the circles on the slide (benches and hydrant on the grass, low railing and drain) are our reading of the dots with the walk-through video, not labels the box knows.
- Small lines in the LiDAR map: the short slanted lines left of the track are the foot of the building with its low railing and drain, seen at an angle; they are too short and slanted to count as an edge line.
- What we measured (replay of the same line detector): the left edge line is found in about 10 percent of the walkway scans in campus 9 and 35 percent in log 8. In between the rover holds the wall direction. A stronger centering, using the nearest object on the left, is the next improvement and is not in version 10.
- Corridor: right railing line 0.35 to 1.3 m, left facade 1.4 to 2.6 m, total width 2.4 to 3.3 m, for 1 second. Found at 35.6 m in campus 9 and 40.0 m in log 8, never earlier.
- Entrance: the facade steps back by 0.2 m or more and the glass drops LiDAR beams. The stop is 1.5 m later.
- Edge assist: in log 8 the rover stood 60 s at the cobble-to-interlock edge. Now it pushes with more power, backs up 0.25 m and pushes again, but only when nothing is closer than 1.2 m ahead, so it never pushes against an obstacle.

- Hold rule: an obstacle that vanishes closer than 0.45 m without being seen moving away is still there. It is released only when several beams see more than 1 m free, or the ultrasonic reads more than 1 m.

Likely questions

Does it keep out of the grass? It tries to: it keeps the distance to the left edge when it sees a line there, otherwise it holds the heading of the wall. In campus 9 the line was found in only about 10 percent of scans, so this is a weak point; the video of the run is the check.

Why did it not stop in campus 9? Two reasons: the corridor rule missed (right railing at 0.9 m, rule wanted 0.85 or less) and the bollard was closer than the LiDAR minimum range.

Will it stop at the right door? Replay says 48.5 m, the entrance is at 47.0 m. It must be checked outdoors.

What if a person stands at the entrance? It holds and finishes the route there. It never drives into or around people.